Statistical Learning & Linear Regression

A Visual Journey Through ISLR Chapters 2–3

Machine Learning with R

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Week 1, 2025

The Fundamental Problem

Given:

- Data: $(x_1, y_1), \ldots, (x_n, y_n)$
- $x_i \in \mathbb{R}^p$ (predictors)
- $y_i \in \mathbb{R}$ (response)

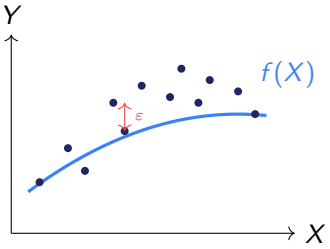
Find:

$$Y = f(X) + \varepsilon$$

f = systematic signal
ε = random noise, $E[\varepsilon] = 0$

Key Question:

What is the **shape** of f?



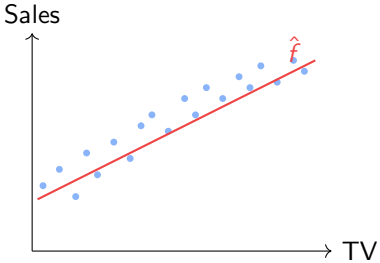
A Real Example: Advertising Data

200 markets observed:

- Y = Sales (thousands)
- X_1 = TV budget ($k)
- X_2 = Radio budget ($k)
- X_3 = Newspaper budget ($k)

The manager asks:

1. Does advertising work?
2. Which media matter?
3. How much should we spend?



Clear upward trend!

Why Estimate f? Two Goals

Prediction

$$\hat{Y} = \hat{f}(X)$$

- f can be a **black box**
- We only care about $\hat{Y}$ accuracy
- Complex models OK

Example: Stock price tomorrow

Inference

$$Y = \beta_0 + \beta_1 X_1 + \cdots$$

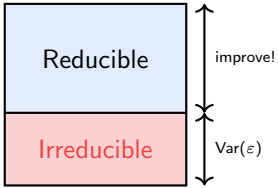
- We want to understand f itself
- **Which** X_j matter?
- **How** does each affect Y?

Example: Which ads drive sales?

Most real problems involve **both** goals
$\Rightarrow$ choose method accordingly

Reducible vs. Irreducible Error

$$E\left[(Y-\hat{Y})^2\right] = \underbrace{[f(X)-\hat{f}(X)]^2}_{\text{reducible}} + \underbrace{\text{Var}(\varepsilon)}_{\text{irreducible}}$$



Reducible: Better model $\Rightarrow$ smaller error
Irreducible: $\text{Var}(\varepsilon)$ is the floor

No model can predict better than $\text{Var}(\varepsilon)$!

Parametric vs. Non-Parametric

Parametric:

Step 1: Assume a form

$$f(X) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

Step 2: Fit (estimate β's)

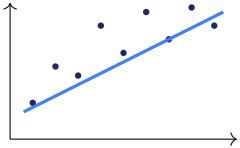
- Simple: estimate $p + 1$ numbers
- May not match true f

Non-Parametric:

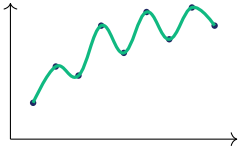
No assumed form for f

Get "close" to data without being too wiggly

- Flexible: many shapes
- Needs **lots** of data
- Overfitting risk

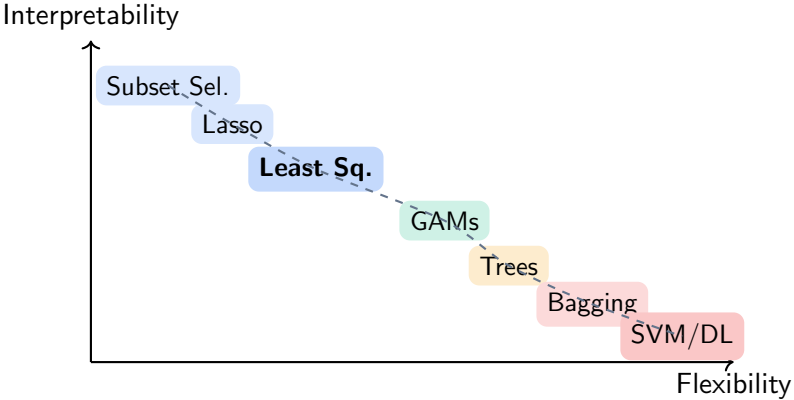


Linear (parametric)



Spline (non-param.)

The Flexibility–Interpretability Trade-Off



More flexible $\neq$ better predictions!

Measuring Fit: Mean Squared Error

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{f}(x_i)\right)^2$$

Training MSE: computed on data used to fit
Test MSE: computed on **new** data

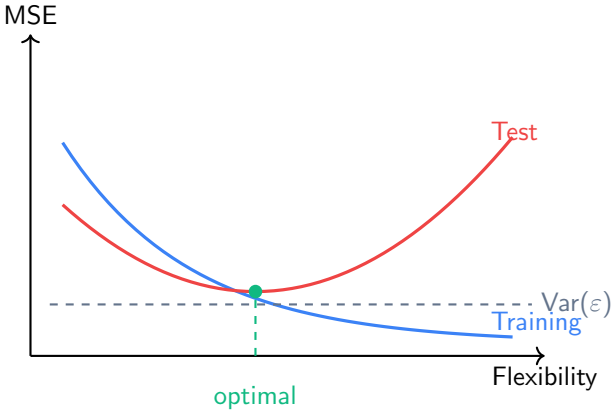
We care about test MSE!

Numerical example:

y_i	$\hat{y}_i$	$(y_i - \hat{y}_i)^2$
3.0	2.8	0.04
5.0	4.5	0.25
7.0	7.2	0.04
4.0	4.8	0.64

$\text{MSE} = \frac{0.04+0.25+0.04+0.64}{4} = 0.24$

Training MSE vs. Test MSE



- Training MSE **always** decreases with flexibility
- Test MSE is **U-shaped** $\Rightarrow$ sweet spot exists
- Past the sweet spot: overfitting

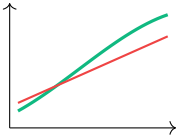
The Bias-Variance Decomposition

$$E\left[(y_0 - \hat{f}(x_0))^2\right] = \underbrace{\mathrm{Var}(\hat{f}(x_0))}_{\text{variance}} + \underbrace{[\mathrm{Bias}(\hat{f}(x_0))]^2}_{\text{bias}^2} + \underbrace{\mathrm{Var}(\varepsilon)}_{\text{irreducible}}$$

Bias = systematic error

Simple model ⇒ high bias

Example: Fitting a line when truth is quadratic

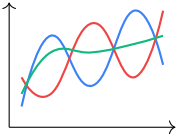


misses curvature

Variance = sensitivity to training data

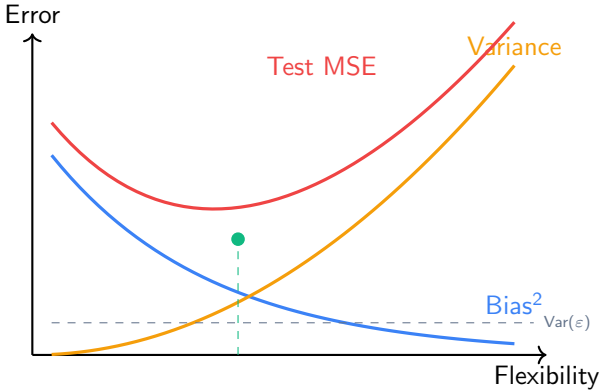
Complex model ⇒ high variance

Example: Different datasets give wildly different curves



3 different training sets

Bias-Variance: The U-Shape

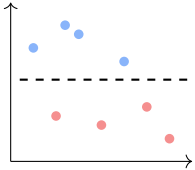


Key insight: We can never observe this decomposition in practice!
$\Rightarrow$ Use **cross-validation** (Chapter 5) to estimate test MSE.

Supervised vs. Unsupervised

Supervised

For each x_i we observe y_i

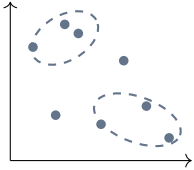


labels known

Regression, Classification

Unsupervised

Only x_i, no y_i



find structure?

PCA, Clustering (Week 10)

Regression vs. Classification

Regression: Y is quantitative

$$Y \in \mathbb{R}$$

Examples: price, salary, mpg, sales

Method: Linear Regression

Classification: Y is categorical

$$Y \in \{0, 1\} \text{ or } \{A, B, C\}$$

Examples: spam/not, cancer type

Method: Logistic Regression (Week 2)

The **predictor type** (qualitative/quantitative) matters less.
Most methods handle both.

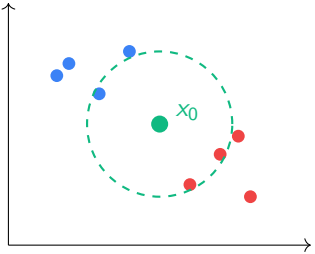
KNN Classifier: Intuition

Algorithm:

1. Pick a new point x_0
2. Find K nearest neighbors
3. Majority vote $\Rightarrow$ class

$$P(Y = j \mid X = x_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} \mathbf{1}(y_i = j)$$

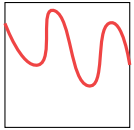
Example: $K = 3$



2 blue, 1 red $\Rightarrow$ **blue!**

KNN: Effect of K

$K = 1$

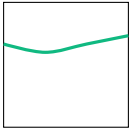


overfits

Training error $= 0$

Very wiggly boundary

$K = 10$



just right

Balanced

Smooth, generalizes well

$K = 100$

underfits

Nearly linear

High bias

$\frac{1}{K}$ controls flexibility: small K = flexible, large K = rigid

Chapter 2: Summary

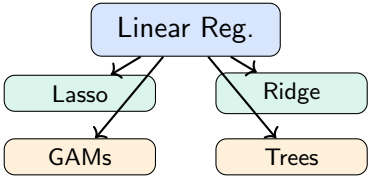
- $Y = f(X) + \varepsilon$
- Prediction vs. Inference
- Parametric vs. Non-parametric
- Bias–Variance trade-off
- MSE: training vs. test
- Flexibility $\leftrightarrow$ Interpretability
- No free lunch!
- KNN: K controls smoothness

Next: Linear Regression — the workhorse of statistical learning

Why Linear Regression?

- Simple, interpretable, widely used
- **Foundation** for Lasso, Ridge, GAMs, ...
- Concepts (RSS, R^2, F-test) recur everywhere
- Benchmark for all fancier methods

"Understand linear regression deeply before studying anything else." — ISLR



Simple Linear Regression

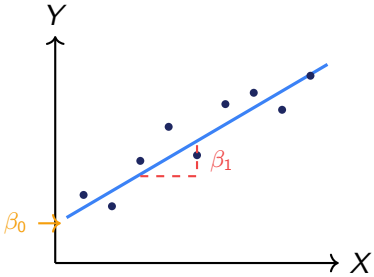
Model:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

β_0 = intercept (value of Y when $X = 0$)
β_1 = slope (change in Y per unit X)

Estimate:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$$

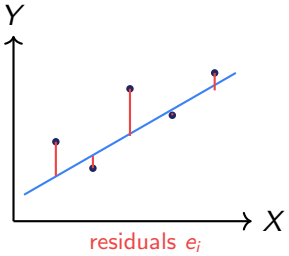


Least Squares: The Idea

Choose $\hat{\beta}_0, \hat{\beta}_1$ to minimize:

$$\text{RSS} = \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

Minimize the sum of squared vertical distances



Closed-form solution:

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

The line always passes through $(\bar{x}, \bar{y})$

Least Squares: Worked Example

Data: 5 points

x_i	y_i
1	2.1
2	3.8
3	5.2
4	6.5
5	8.1

$\bar{x} = 3, \quad \bar{y} = 5.14$

Computation:

$\sum(x_i - \bar{x})(y_i - \bar{y}) = 11.9$

$\sum(x_i - \bar{x})^2 = 10$

$$\hat{\beta}_1 = \frac{11.9}{10} = 1.19$$

$$\hat{\beta}_0 = 5.14 - 1.19 \times 3 = 1.57$$

$\hat{Y} = 1.57 + 1.19X$

Each unit of $X \Rightarrow Y$ increases by 1.19

How Good Is $\hat{\beta}_1$? Standard Errors

Different samples $\Rightarrow$ different $\hat{\beta}_1$

Standard Error quantifies this variability:

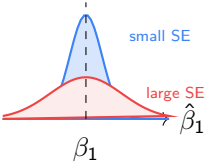
$$\mathrm{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}$$

$$\mathrm{SE}(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_i (x_i - \bar{x})^2} \right]$$

where $\sigma^2 \approx \mathrm{RSE}^2 = \frac{\mathrm{RSS}}{n-2}$

SE is smaller when:

- n is larger
- x_i's are more spread out
- σ^2 (noise) is smaller



Hypothesis Test: Is X Related to Y?

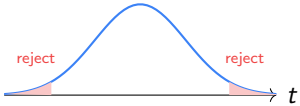
$$H_0 : \beta_1 = 0 \quad \text{(no relationship)}$$

$$H_a : \beta_1 \neq 0 \quad \text{(some relationship)}$$

Test statistic:

$$t = \frac{\hat{\beta}_1}{\text{SE}(\hat{\beta}_1)} \sim t_{n-2} \quad \text{under } H_0$$

Example:
$\hat{\beta}_1 = 0.0475$, SE $= 0.0027$
$t = 0.0475/0.0027 = 17.67$
p-value < 0.0001
Reject H_0! Strong evidence.



p-value = shaded area

Confidence Intervals

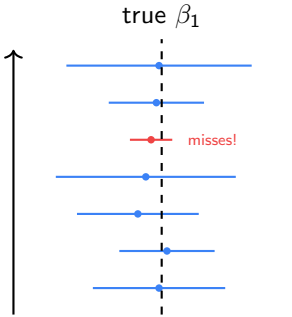
95% CI for β_1:

$$\hat{\beta}_1 \pm 2 \cdot \mathrm{SE}(\hat{\beta}_1)$$

Interpretation:
In repeated sampling, 95% of such intervals contain the true β_1

Example:
$\hat{\beta}_1 = 0.0475 \pm 2(0.0027)$
$= [0.0421,\ 0.0529]$

Does not contain 0 $\Rightarrow$ significant



95% of CIs capture truth

Model Accuracy: RSE and R^2

RSE (Residual Standard Error):

$$\text{RSE} = \sqrt{\frac{\text{RSS}}{n-2}}$$

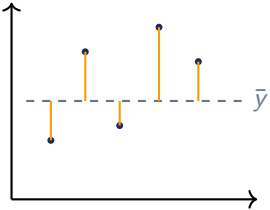
- In units of Y
- "Typical prediction error"
- *Advertising:* RSE $= 3.26$
- Average sales $= 14 \Rightarrow 23\%$ error

R^2 (Proportion of variance explained):

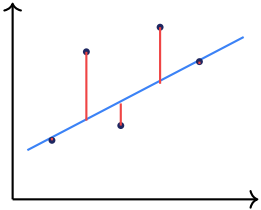
$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

- $R^2 \in [0, 1]$
- $R^2 = 0$: model $= \bar{y}$
- $R^2 = 1$: perfect fit
- In SLR: $R^2 = r_{XY}^2$

R^2: Visual Intuition



$$\text{TSS} = \sum (y_i - \bar{y})^2$$
total variability



$$\text{RSS} = \sum (y_i - \hat{y}_i)^2$$
leftover variability

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} = \frac{\text{variability explained by model}}{\text{total variability}}$$

Multiple Linear Regression

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon$$

Interpretation of β_j:
Average change in Y per unit change in X_j, **holding all other X_k fixed**

Advertising:

$$\begin{aligned}\text{Sales} = 2.94 &+ 0.046 \cdot \text{TV} \\ &+ 0.189 \cdot \text{Radio} \\ &- 0.001 \cdot \text{News}\end{aligned}$$

Key results:

	$\hat{\beta}$	p
TV	0.046	< 0.001
Radio	0.189	< 0.001
News	-0.001	0.86

Newspaper not significant!
(Confounded with Radio: $r = 0.35$)

Why Newspaper Vanishes

Simple regression:

	p-value
TV alone	< 0.001
Radio alone	< 0.001
News alone	< 0.001

Newspaper seems important!

Multiple regression:

	p-value
TV	< 0.001
Radio	< 0.001
Newspaper	0.86

Not significant!

$$\text{Newspaper} \xleftrightarrow{r=0.35} \text{Radio} \xrightarrow{\text{drives}} \text{Sales}$$

Newspaper is a **proxy** *for Radio, not an independent driver.*

The F-Test: Overall Significance

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0$$

$$F = \frac{(\text{TSS} - \text{RSS})/p}{\text{RSS}/(n - p - 1)}$$

Intuition:

- Numerator: improvement per predictor
- Denominator: residual noise per df
- Under H_0: $F \approx 1$
- Large $F \Rightarrow$ predictors help!

Why not just t-tests?

With $p = 100$ predictors and all $\beta_j = 0$:

Expect ~ 5 false positives at $\alpha = 0.05$!

F-test avoids this **multiple testing** problem.

Qualitative Predictors: Dummy Variables

Problem: How to encode "Region"?

Create **indicator** (dummy) variables:

$$x_i = \begin{cases} 1 & \text{if female} \\ 0 & \text{if male} \end{cases}$$

k levels $\Rightarrow k - 1$ dummies

R does this automatically!

```
lm(y ~ factor(region))
```

Example: 3 ethnicities

	x_1	x_2
African Am.	0	0
Asian	1	0
Caucasian	0	1

β_0 = mean for baseline
β_1 = Asian − baseline
β_2 = Caucasian − baseline

Interaction Terms: Synergy

Without interaction:

$$\text{Sales} = \beta_0 + \beta_1 \cdot \text{TV} + \beta_2 \cdot \text{Radio}$$

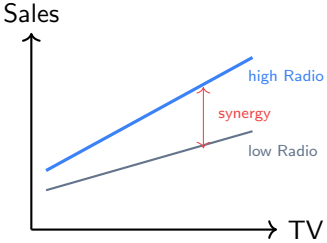
Effect of TV is **constant** regardless of Radio

With interaction:

$$\text{Sales} = \beta_0 + \beta_1 \cdot \text{TV} + \beta_2 \cdot \text{Radio} + \beta_3 \cdot \text{TV} \times \text{Radio}$$

Effect of TV $= \beta_1 + \beta_3 \cdot \text{Radio}$

More Radio $\Rightarrow$ TV is more effective!



R: `lm(sales ~ TV * radio)`

Polynomial Regression

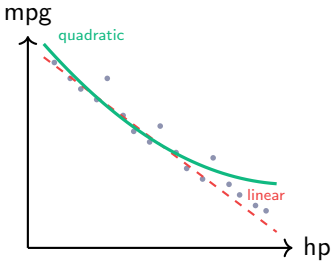
$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon$$

Still **linear** in the parameters!

Auto data:

Model	R^2
mpg $\sim$ hp	0.606
mpg $\sim$ hp + hp^2	0.688
mpg $\sim$ poly(hp, 5)	0.693

Quadratic captures most gain!



Potential Problems: Overview

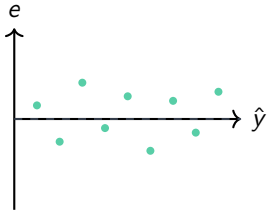
#	Problem	Diagnostic	Fix
1	Non-linearity	Residual plot	Polynomial/log
2	Non-constant var.	Scale-Location	Transform Y
3	Correlated errors	Residuals vs. time	Time series
4	Outliers	Studentized resid.	Investigate
5	High leverage	Hat values h_{ii}	Investigate
6	Collinearity	VIF	Drop/combine

Always check residual plots before trusting results!

Call to action: `par(mfrow=c(2,2)); plot(fit)`

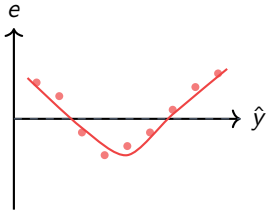
Problem 1: Non-Linearity

Good: Random scatter



Assumptions satisfied!

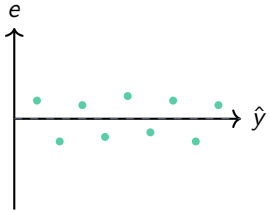
Bad: Curved pattern



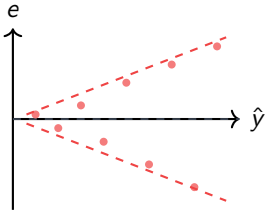
Non-linearity! Add X^2 term.

Problem 2: Heteroscedasticity

Good: Constant spread



Bad: Funnel shape



Variance grows! Try $\log(Y)$.

Fix: Transform response: $\log(Y)$, $\sqrt{Y}$, or use weighted least squares

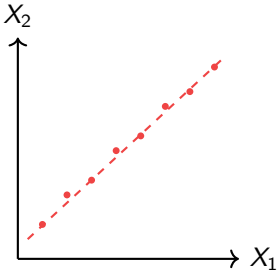
Problem 6: Collinearity

Two predictors highly correlated $\Rightarrow$ hard to separate their effects

Variance Inflation Factor:

$$\mathrm{VIF}(\hat{\beta}_j) = \frac{1}{1 - R^2_{X_j|X_{-j}}}$$

VIF	Interpretation
1	No collinearity
1–5	Moderate
5–10	High — investigate
> 10	Severe — fix it!



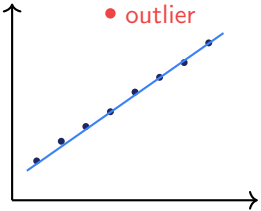
$r \approx 0.99$

Can't tell effects apart!

R: `car::vif(fit)`

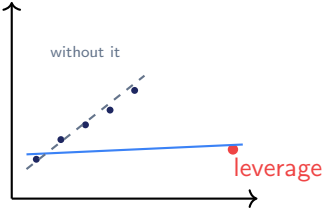
Outliers vs. High Leverage

Outlier: unusual y_i



Little effect on slope, but inflates RSE

High leverage: unusual x_i



Pulls the line!

Detect: $h_{ii} \gg (p+1)/n$

R^2 Always Increases — Adjusted R^2 Fixes This

$$\text{Adj. } R^2 = 1 - \frac{\text{RSS}/(n-p-1)}{\text{TSS}/(n-1)}$$

Example:

Model	R^2	Adj. R^2
TV	0.612	0.610
TV + Radio	0.897	0.896
TV + Radio + News	0.897	0.896

Adding News: R^2 unchanged!

Key insight:

R^2 **never** decreases when you add a predictor (even a useless one)

Adj. R^2 **penalizes** for extra parameters $\Rightarrow$ can decrease

Better for comparing models

Confidence vs. Prediction Intervals

Confidence interval:

For $E[Y \mid X = x_0]$

Where the **average** response lies

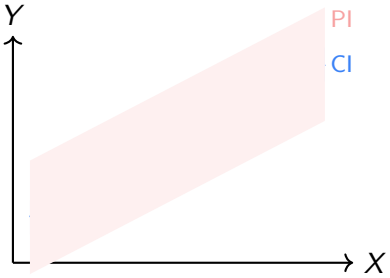
Narrower

Prediction interval:

For a **new individual** Y at x_0

Includes $\mathrm{Var}(\varepsilon)$

Always wider!



OLS in Matrix Form

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \varepsilon$$

where $\mathbf{X}$ is $n \times (p+1)$:

$$\mathbf{X} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1p} \\ 1 & x_{21} & \cdots & x_{2p} \\ \vdots & & & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{pmatrix}$$

Solution:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

Properties:

- Unbiased: $E[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$
- BLUE (Gauss-Markov)
- $\text{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$

The Hat Matrix: Who Influences Whom?

$$\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}, \qquad \mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$$

h_{ii} = leverage of observation i

- $\frac{1}{n} \leq h_{ii} \leq 1$
- Average: $\frac{p+1}{n}$
- Large $h_{ii} \Rightarrow$ influential

Cook's distance combines leverage *and* residual:

$$D_i = \frac{e_i^2}{p \cdot \text{MSE}} \cdot \frac{h_{ii}}{(1 - h_{ii})^2}$$

$D_i > 1 \Rightarrow$ influential point

Numerical example:

$n = 50$, $p = 3$

Average leverage $= 4/50 = 0.08$

Flag if $h_{ii} > 3 \times 0.08 = 0.24$

R: `hatvalues(fit)`

R: `cooks.distance(fit)`

The 4 Diagnostic Plots in R

```
par(mfrow = c(2,2)); plot(fit)
```

1. **Residuals vs Fitted**
   - Checks linearity
   - Look for curves
2. **Normal Q-Q**
   - Checks normality
   - Points on diagonal = good
3. **Scale-Location**
   - Checks constant variance
   - Horizontal band = good
4. **Residuals vs Leverage**
   - Finds influential points
   - Watch Cook's distance lines

Call to action: Open `Lab_Week1.R`, run lines 85–115

KNN Regression vs. Linear Regression

KNN regression:

$$\hat{f}(x_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} y_i$$

Average of K nearest neighbors

- No assumptions
- Curse of dimensionality!

When does each win?

	Linear	KNN
Linear f	✓	
Non-linear f		✓
Large p	✓	
Small p		✓

In high dimensions, even KNN struggles. Structure helps.

Curse of Dimensionality: A Numerical Example

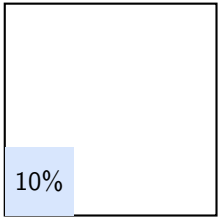
To capture 10% of data in KNN, how far must we reach?

Side length of hypercube:

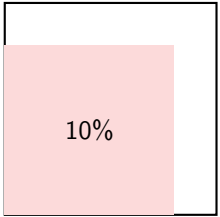
$$\ell = 0.1^{1/p}$$

p	ℓ
1	0.10
2	0.32
5	0.63
10	0.80
100	0.98

At $p = 10$: need 80% of range!



$p = 2$: local



$p = 10$: global!

R Functions: Quick Reference

`lm(y ~ x1 + x2, data)`	Fit linear model
`summary(fit)`	Coefficients, R^2, F-test
`confint(fit)`	95% CIs
`predict(fit, newdata)`	Predictions
`plot(fit)`	4 diagnostic plots
`car::vif(fit)`	VIF for collinearity
`anova(fit1, fit2)`	Compare nested models

`y ~ x1 * x2`	Main effects + interaction
`y ~ poly(x, 2)`	Quadratic regression
`y ~ log(x)`	Log transform
`y ~ . - z`	All predictors except z

Key Formulas: Cheat Sheet

Estimation:

$$\hat{\beta}_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

Quality:

$$R^2 = 1 - \text{RSS}/\text{TSS}$$

$$\text{RSE} = \sqrt{\text{RSS}/(n-p-1)}$$

Inference:

$$t = \hat{\beta}_j / \text{SE}(\hat{\beta}_j)$$

$$F = \frac{(\text{TSS} - \text{RSS})/p}{\text{RSS}/(n-p-1)}$$

$$\text{CI} : \hat{\beta}_j \pm t_{\alpha/2} \cdot \text{SE}$$

Diagnostics:

$$\text{VIF}_j = 1/(1 - R^2_{X_j|X_{-j}})$$

$$h_{ii} = [\mathbf{H}]_{ii}$$

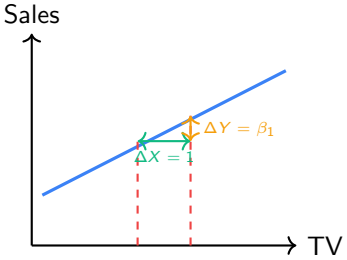
Think About It: What Does $\beta_1 = 0.046$ Mean?

Advertising data:
$\text{Sales} = 7.03 + 0.0475 \cdot \text{TV}$

Step by step:

- TV budget goes from \$100k to \$101k
- $\Delta\text{TV} = 1$ (thousand dollars)
- $\Delta\text{Sales} = 0.0475 \times 1 = 0.0475$
- That's 47.5 extra units sold

Worth it? Depends on profit per unit!



The Intercept: Often Meaningless

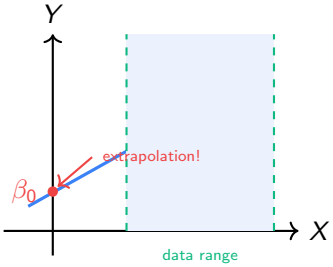
$\hat{\beta}_0 = 7.03$ in Sales $\sim$ TV

Interpretation:
Expected Sales when TV $= 0$

Problem: Does TV $= 0$ make sense?

- Sometimes yes (no advertising)
- Often extrapolation beyond data
- `mpg` at hp $= 0$? Nonsense!

Focus on the slope, not the intercept



Worked Example: Computing R^2

y_i	$\hat{y}_i$	$(y_i - \bar{y})^2$	$(y_i - \hat{y}_i)^2$
2	2.3	4	0.09
4	3.8	0	0.04
5	4.5	1	0.25
3	3.0	1	0.00
6	5.3	4	0.49

$\bar{y} = 4$

$$\text{TSS} = 4 + 0 + 1 + 1 + 4 = 10$$
$$\text{RSS} = 0.09 + 0.04 + 0.25 + 0 + 0.49 = 0.87$$

$$R^2 = 1 - \frac{0.87}{10} = 0.913$$

91.3% of variability explained!

$$\text{RSE} = \sqrt{0.87/3} = 0.54$$

R^2: What's a "Good" Value?

It depends on the field!

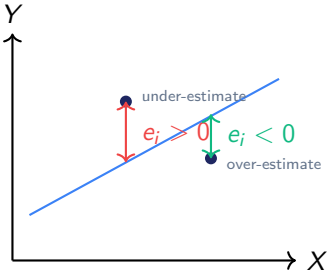
Domain	Typical R^2
Physics experiments	> 0.99
Engineering	0.8–0.95
Social sciences	0.3–0.6
Finance (daily)	0.01–0.05

$R^2 = 0.05$ can be **valuable** in trading!

Common mistakes:

- R^2 close to 1 does NOT mean model is correct
- High R^2 can come from overfitting
- Low R^2 doesn't mean X is irrelevant
- Always pair with residual diagnostics

Residual = Observed − Predicted



$$e_i = y_i - \hat{y}_i$$

Key properties:

- $\sum e_i = 0$ (by construction)
- $\sum x_i e_i = 0$
- $\text{RSS} = \sum e_i^2$

Residuals tell us what the model missed!

From Correlation to Regression

Correlation:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \cdot \sum(y_i - \bar{y})^2}}$$

$r \in [-1, 1]$
Measures *strength* and *direction*

Regression slope:

$$\hat{\beta}_1 = r \cdot \frac{s_Y}{s_X}$$

Same sign as r, but scaled!

$r = 0.78$ (TV vs Sales)
$\hat{\beta}_1 = 0.0475$ (in original units)

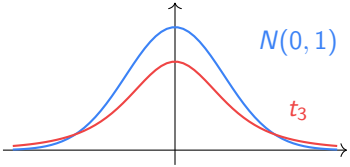
In SLR: $R^2 = r^2 = 0.78^2 = 0.61$ ✓

The t-Distribution: Why Not Normal?

We estimate σ^2 with RSE^2
This adds uncertainty $\Rightarrow$ heavier tails

$$t = \frac{\hat{\beta}_1}{\text{SE}(\hat{\beta}_1)} \sim t_{n-2}$$

For large n: $t_{n-2} \approx N(0,1)$
For small n: t_{n-2} has fatter tails
$\Rightarrow$ wider CIs, harder to reject H_0



heavier tails = more uncertainty

Worked Example: Hypothesis Test

Advertising: Is TV related to Sales?

$\hat{\beta}_1 = 0.0475$
$\text{SE}(\hat{\beta}_1) = 0.0027$

$$t = \frac{0.0475}{0.0027} = 17.67$$

$n - 2 = 198$ degrees of freedom
p-value $< 2 \times 10^{-16}$

Decision:
$|t| = 17.67 \gg 1.97$ (critical value)

Reject H_0

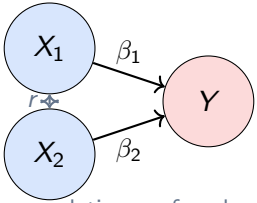
Extremely strong evidence that TV budget is related to Sales

(The chance of seeing $t = 17.67$ if $\beta_1 = 0$ is essentially zero)

Multiple Regression: Why Not Just Run Separate SLRs?

Problem:
Running mpg $\sim$ horsepower and mpg $\sim$ weight separately
$\Rightarrow$ **cannot** hold other variables fixed
$\Rightarrow$ confounding!

MLR advantage:
Each $\hat{\beta}_j$ is the **partial effect** of X_j after removing the effect of all other predictors



correlation confounds separate SLRs

F-Test: Worked Example

Advertising: Sales $\sim$ TV + Radio + Newspaper

$n = 200$, $p = 3$
TSS $= 5417.15$
RSS $= 556.91$

$$F = \frac{(5417.15 - 556.91)/3}{556.91/(200 - 3 - 1)}$$

$$= \frac{1620.1}{2.84} = 570.3$$

Under H_0: $F \sim F_{3,196}$

$F = 570.3 \gg 1$
p-value $< 10^{-16}$

At least one predictor matters!

(But which ones? Check individual t-tests)

Interactions: Quantitative $\times$ Qualitative

Without interaction:

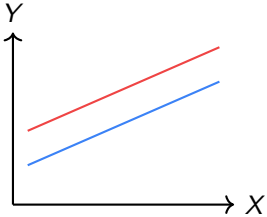
$Y = \beta_0 + \beta_1 X + \beta_2 D$

Same slope, different intercepts

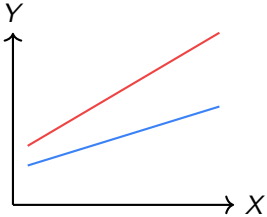
With interaction:

$Y = \beta_0 + \beta_1 X + \beta_2 D + \beta_3 (X \cdot D)$

Different slopes AND intercepts



without interaction
parallel lines



with interaction
different slopes

The Hierarchical Principle

Rule: If an interaction $X_1 \times X_2$ is in the model, keep both main effects X_1 and X_2.

Correct:
$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2$

Wrong:
$Y = \beta_0 + \beta_3 X_1 X_2$

Why?

- Removing X_1 changes meaning of β_3
- Interaction without main effects is hard to interpret
- Even if β_1 is not significant, keep it

Log Transformations: Three Scenarios

Model	Formula	Interpretation of β_1
Level-level	$Y = \beta_0 + \beta_1 X$	$\Delta X = 1 \Rightarrow \Delta Y = \beta_1$
Log-level	$\log Y = \beta_0 + \beta_1 X$	$\Delta X = 1 \Rightarrow \%\Delta Y \approx 100\beta_1$
Level-log	$Y = \beta_0 + \beta_1 \log X$	$\%\Delta X = 1 \Rightarrow \Delta Y \approx \beta_1/100$
Log-log	$\log Y = \beta_0 + \beta_1 \log X$	**Elasticity:** $\%\Delta X \Rightarrow \%\Delta Y$

When to use log**:**

- Skewed response (income, prices)
- Funnel-shaped residuals (heteroscedasticity)
- Multiplicative relationships

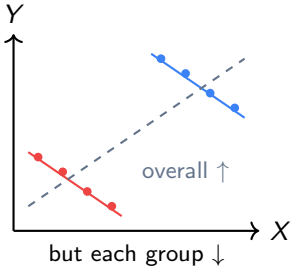
Simpson's Paradox: A Warning

The trap:
A trend in subgroups can **reverse** when groups are combined!

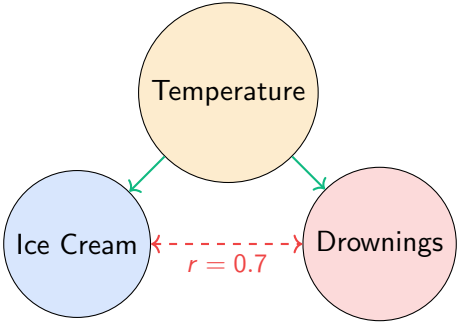
Example:
Treatment appears harmful overall, but is beneficial within each group

Cause: Confounding variable (group membership)

Lesson: Always consider omitted variables



Correlation $\neq$ Causation



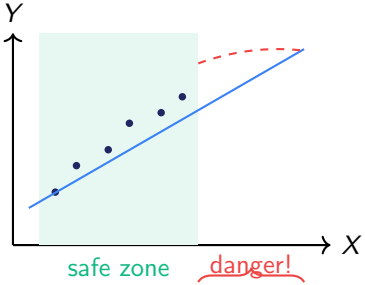
- Regression finds **associations**, not causes
- Confounders create spurious correlations
- Causation requires: experiments, randomization, or careful causal inference

Extrapolation: Stay Within the Data Range

Dangerous:
Predicting outside the range of observed X values

The linear relationship may not hold beyond the data

Example:
TV budget in data: \$0–\$300k
Predicting at \$500k?
Unreliable!



Weighted Least Squares: When Variance Varies

Standard OLS:
Assumes $\text{Var}(\varepsilon_i) = \sigma^2$ for all i

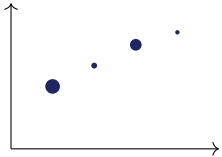
WLS:
Allows $\text{Var}(\varepsilon_i) = \sigma^2/w_i$
Minimize:

$$\sum_{i=1}^{n} w_i(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

R: `lm(y ~ x, weights = w)`

Intuition:
Give **more weight** to observations with smaller variance

Common case:
y_i is an average of n_i observations
$\Rightarrow w_i = n_i$



bigger dot = larger w_i

Model Selection Preview: Which Predictors to Keep?

With p predictors: 2^p possible models!

p	# Models
5	32
10	1,024
20	1,048,576

Can't check all! Need smart strategies.

Preview (Chapter 6):

- Forward selection
- Backward selection
- Best subset selection
- Lasso (automatic!)

Criteria:
AIC, BIC, Adj. R^2, C_p, CV

Reading the R Output: A Guided Tour

```
> summary(lm(sales ~ TV))
```

	Est.	SE	t	p
Int	7.03	0.46	15.4	<0.001
TV	0.048	0.003	17.7	<0.001

RSE: 3.26 on 198 df
R^2: 0.612
Adj R^2: 0.610
F: 312.1, $p < 10^{-16}$

Read it like a story:

1. F-test: overall model significant?
   Yes! ($p < 10^{-16}$)
2. Coefficients: which matter?
   TV significant ($p < 0.001$)
3. R^2: how much explained?
   61% of variance
4. RSE: how big are errors?
   $\pm$3.26 thousand units

VIF: A Worked Example

Auto data: mpg $\sim$ cylinders + displacement + weight

Step 1: Regress displacement on cylinders and weight
$R^2_{\text{disp}|\text{others}} = 0.89$

Step 2: Compute VIF

$$\text{VIF}_{\text{disp}} = \frac{1}{1 - 0.89} = 9.1$$

VIF $> 5 \Rightarrow$ problematic collinearity!

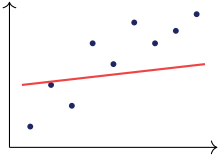
Typical Auto VIFs:

Predictor	VIF
cylinders	9.1
displacement	14.3
horsepower	8.7
weight	10.8
year	1.2
origin	1.8

year & origin: no issue!

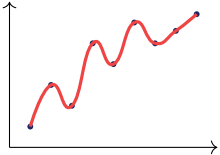
Overfitting: The Core Danger

Underfitting:



too simple
high bias

Overfitting:



too complex
high variance

Golden rule: Never evaluate a model on the data used to train it!

Next lecture topic: Cross-Validation (Chapter 5) formalizes this

Advertising: The Full Story

Model comparison:

Model	R^2
Sales $\sim$ TV	0.612
Sales $\sim$ TV + Radio	0.897
Sales $\sim$ TV + Radio + News	0.897
Sales $\sim$ TV $\times$ Radio	0.968

Adding Radio: huge jump!
Adding Newspaper: nothing
Adding interaction: best model

Marketing recommendations:

1. Invest in **TV** and **Radio**
2. Drop **Newspaper** budget
3. Plan TV and Radio **jointly** (synergy effect!)
4. Don't extrapolate beyond observed budget ranges

Auto Data: Linear vs. Quadratic in Detail

Linear: mpg $\sim$ hp

Metric	Value
R^2	0.606
RSE	4.91
$\hat{\beta}_1$	-0.158

Residual plot: U-shape!

Quadratic: mpg $\sim$ hp + hp^2

Metric	Value
R^2	0.688
RSE	4.37
$\hat{\beta}_2$	0.0004

Residual plot: much better!

ANOVA: $F = 51.4$, $p < 10^{-11} \Rightarrow$ quadratic term highly significant

`anova(linear_fit, quad_fit)`

What Goes Into predict() ?

`predict(fit, newdata, interval)`

Argument	What
`fit`	your `lm()` object
`newdata`	data.frame with X
`interval`	"confidence" or "prediction"

Example:

```
new <- data.frame(hp = 98)
predict(fit, new,
        interval = "confidence")
```

Returns: `fit`, `lwr`, `upr`

	CI	PI
fit	24.47	24.47
lwr	23.97	14.81
upr	24.96	34.12

Assumptions Checklist: Before You Trust Results

1. Linearity? $\Rightarrow$ Residuals vs Fitted
2. Independence? $\Rightarrow$ Residuals vs Time order
3. Constant variance? $\Rightarrow$ Scale-Location plot
4. Normality? $\Rightarrow$ Q-Q plot
5. No multicollinearity? $\Rightarrow$ `car::vif(fit)`

If assumptions are violated, inference (p-values, CIs) may be wrong!

Gauss-Markov: Why OLS Is Special

Theorem:
Among all **linear** and **unbiased** estimators of β, OLS has the **smallest variance**.

$\hat{\beta}_{\text{OLS}}$ is **BLUE**:

- **B**est
- **L**inear
- **U**nbiased
- **E**stimator

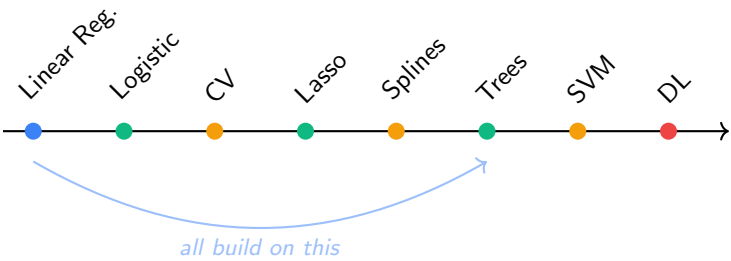
Requirements:

1. $E[\varepsilon_i] = 0$
2. $\text{Var}(\varepsilon_i) = \sigma^2$
3. $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$

Note:

- Does NOT require normality!
- Normality needed only for exact t and F distributions

Beyond Week 1: The Road Ahead



Linear regression limitations we'll address:

- Non-linearity ⇒ Splines, Trees, Neural Nets
- Too many predictors ⇒ Lasso, Ridge
- Model evaluation ⇒ Cross-Validation
- Classification ⇒ Logistic Regression, SVM

Exercise 1: Auto Dataset (ISLR 3.7 Q8)

Data: `Auto` from ISLR2 $n = 392,\ p = 8$

1. Fit SLR: `lm(mpg ~ horsepower)`
2. Interpret $\hat{\beta}_1$ in words
3. Predict mpg at horsepower $= 98$
4. Compute 95% CI and PI at hp $= 98$
5. Plot data with regression line
6. Check diagnostic plots — any problems?

Hint: Expect $R^2 \approx 0.61$ and a U-shaped residual plot

Exercise 2: Multiple Regression (ISLR 3.7 Q9)

Data: `Auto` Fit: `mpg ~ . - name`

1. Produce a scatterplot matrix
2. Compute correlation matrix
3. Fit MLR with all predictors
4. Which predictors are significant?
5. What does the `year` coefficient mean?
6. Check residual plots
7. Try interaction terms — which are significant?
8. Try log, $\sqrt{X}$, X^2 — do they improve fit?

Exercise 3: Carseats (ISLR 3.7 Q10)

Data: `Carseats` $n = 400$

Fit: `Sales ~ Price + Urban + US`

1. Interpret each coefficient
2. For which can you reject H_0?
3. Fit reduced model (significant predictors only)
4. Compare R^2, Adj. R^2, RSE
5. 95% CIs for the reduced model

Key finding: Urban is not significant ($p \approx 0.94$)

Exercise 4: Boston EDA (ISLR 2.4 Q10)

Data: `Boston` $n = 506$ census tracts

1. How many rows/columns?
2. Pairwise scatterplots for selected variables
3. Which predictors associate with crime rate?
4. Tracts with extreme crime, tax, ptratio?
5. How many tracts bound the Charles River?
6. Median pupil-teacher ratio?
7. Tract with lowest median home value — comment

Hint: `cor(Boston)[, "crim"]` reveals strongest associations

Solution: Exercise 1 — Key Results

$\hat{\beta}_1 = -0.158$, $p < 10^{-16}$

Each unit hp $\Rightarrow$ mpg drops by 0.16

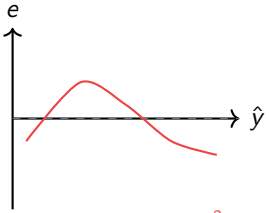
At hp $= 98$: $\hat{Y} = 24.47$

95% CI: [23.97, 24.96]

95% PI: [14.81, 34.12]

$R^2 = 0.606$, RSE $= 4.91$

Residual plot:



U-shape! Add hp^2

With quadratic: $R^2 = 0.688$

Solution: Exercise 2 — Key Results

Significant predictors:
displacement, weight, year, origin

Not significant:
horsepower (collinear with weight)

Year coefficient ≈ 0.75:
Each model year $\Rightarrow +0.75$ mpg
(technology improves efficiency)

Improvements:

Model	R^2
All linear	0.82
+ interactions	0.86
+ log terms	0.87

displacement $\times$ weight interaction is significant

Solution: Exercise 3 — Carseats

	$\hat{\beta}$	p
Price	-0.054	< 0.001
Urban	-0.022	0.936
US	$+1.201$	< 0.001

Reduced model:

```
Sales ~ Price + US
```

Adj. R^2 slightly improves!

Interpretation:

- \$1 more $\Rightarrow$ 0.054 fewer units sold
- Urban/Rural: no difference
- US stores sell 1.2 more units

Drop Urban, keep Price + US

Solution: Exercise 4 — Boston

- $n = 506$, $p = 13$
- Crime correlates with:
  rad ($r = 0.63$), tax ($r = 0.58$)
- 35 tracts bound Charles River
- Median ptratio $= 19.05$

Lowest medv tract:

Variable	Value
crim	38.35
age	100
lstat	30.59
medv	\$5k

Most disadvantaged area

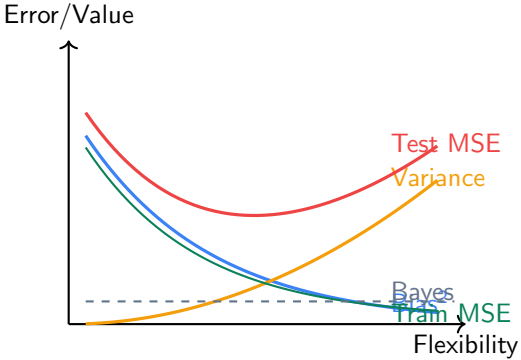
Conceptual Solutions (ISLR 2.4 Q1)

Flexible or inflexible method?

Scenario	Best	Why
(a) Large n, small p	Flexible	Enough data to avoid overfit
(b) Small n, large p	Inflexible	High overfit risk
(c) Non-linear f	Flexible	Rigid model has high bias
(d) High Var(ε)	Inflexible	Flexible follows noise

Always think: bias–variance trade-off!

Conceptual: Sketch the Curves (ISLR 2.4 Q3)



5 curves: Bias^2 (decreasing), Variance (increasing), Test MSE (U-shape), Training MSE (decreasing), Bayes error (constant horizontal line)

Conceptual: Training RSS — Linear vs. Cubic (ISLR 3.7 Q4)

(a) True f is linear:

Training RSS:
Cubic $\leq$ Linear

More flexible always fits training data at least as well

(b) Test RSS:
Linear $\leq$ Cubic (probably)

Cubic overfits — estimates noise as signal

(c) True f is non-linear:

Training RSS:
Cubic $\leq$ Linear (always)

(d) Test RSS:
Not enough information!

Depends on how far from linear:

- Slightly non-linear: linear may win
- Very non-linear: cubic wins

Regression vs. Classification: Practice (ISLR 2.4 Q2)

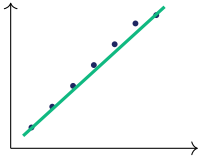
Scenario	Type	n	p	Goal
(a) CEO salary	Regression	500	3	Inference
(b) Product launch	Classification	20	13	Prediction
(c) Exchange rate	Regression	52	3	Prediction

Key distinctions:

- CEO salary is continuous $\Rightarrow$ regression + inference ("which factors?")
- Success/failure is binary $\Rightarrow$ classification + prediction
- % change is continuous $\Rightarrow$ regression + prediction

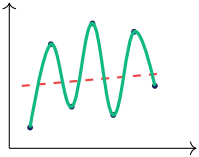
The "No Free Lunch" Theorem — Visually

Dataset A:



Linear wins!

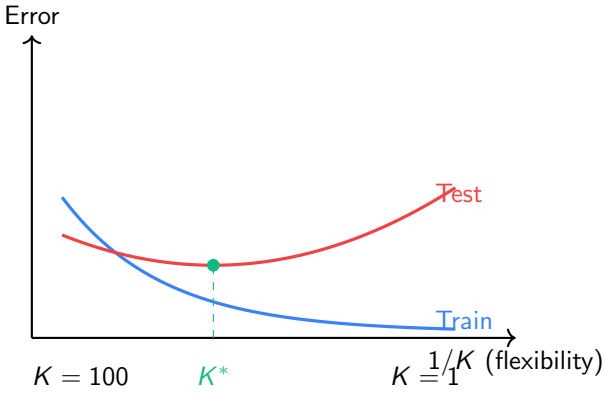
Dataset B:



KNN wins!

No single method is best for all datasets.
The best method depends on the data structure.

Bias-Variance for KNN: K Controls Everything



- At $K = 1$: Training error $= 0$, but test error is high (overfit)
- At $K = n$: Every prediction $= \bar{y}$ (underfit)
- Optimal K^* minimizes test error

How Many Predictors Is Too Many?

Rule of thumb:
Need $\approx$ 10–20 observations per predictor

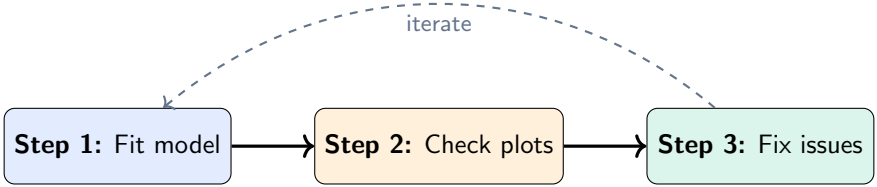
n	p	Ratio
200	3	67 ✓
100	10	10 ∼
50	20	2.5 ×
30	50	0.6 ××

When $p > n$:

- OLS doesn't work! ($\mathbf{X}^T\mathbf{X}$ singular)
- $R^2 = 1$ trivially
- Need regularization (Ch 6)
- Lasso / Ridge to the rescue

High dimensions: Chapter 6 will save you!

Residual Analysis: A 3-Step Workflow




```
fit <- lm(...)
summary(fit)
```

```
plot(fit)
car::vif(fit)
```

Add X^2, $\log(Y)$
Remove collinear X

Regression is iterative, not one-shot!

Comparing Nested Models: The ANOVA Approach

Model 1: mpg $\sim$ horsepower (reduced)
Model 2: mpg $\sim$ horsepower + horsepower^2 (full)

$$F = \frac{(\text{RSS}_1 - \text{RSS}_2)/(\text{df}_1 - \text{df}_2)}{\text{RSS}_2/\text{df}_2}$$

Numbers:
$\text{RSS}_1 = 9386$, $\text{df}_1 = 390$
$\text{RSS}_2 = 7442$, $\text{df}_2 = 389$

$$F = \frac{(9386 - 7442)/1}{7442/389} = 101.6$$

p-value $< 10^{-11}$

Quadratic term is highly significant

R: `anova(fit1, fit2)`
Tests whether the extra predictor(s) provide a significant improvement

Quick Quiz: Test Yourself!

1. What does $R^2 = 0.75$ mean?

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   PI includes $\text{Var}(\varepsilon)$ on top of coefficient uncertainty

Quick Quiz: More Questions

5. Residual plot shows U-shape. What's wrong?

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   R^2: never. Adjusted R^2: yes, if the predictor is useless

7. $\hat{\beta}_1 = -0.05$, SE $= 0.08$. Is this significant?

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7. $\hat{\beta}_1 = -0.05$, SE $= 0.08$. Is this significant?

   $t = -0.05/0.08 = -0.625$. No — $|t| < 2$, p is large

8. When does KNN beat linear regression?

Quick Quiz: More Questions

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   $t = -0.05/0.08 = -0.625$. No — $|t| < 2$, p is large

8. When does KNN beat linear regression?

   When f is highly non-linear AND p is small

The Advertising Correlation Matrix

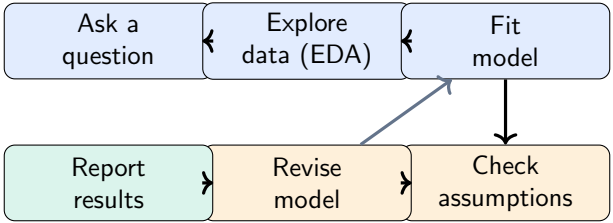
	TV	Radio	News	Sales
TV	1.00	0.05	0.06	**0.78**
Radio	0.05	1.00	0.35	**0.58**
News	0.06	0.35	1.00	0.23
Sales	**0.78**	**0.58**	0.23	1.00

Key observations:

- TV strongest predictor
- Radio–News: $r = 0.35$
  This is the confounder!
- TV & Radio are nearly uncorrelated ($r = 0.05$)

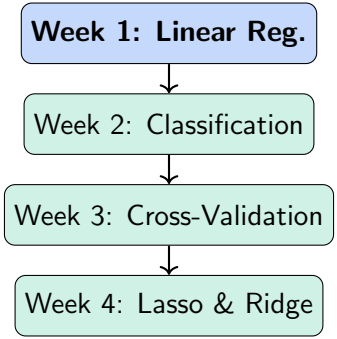
R: `round(cor(Advertising[, -5]), 2)`

Key Takeaway: The Regression Workflow



Regression is a conversation with data, not a single command.

What's Next?



Homework: ISLR Chapter 2: Q1, 2, 3, 10 — Chapter 3: Q8, 9, 10
Lab: `Lab_Week1.R` from `github.com/endri81/ISLRv2-solutions`

Thank You

Questions?

Dr. Endri Raço — Machine Learning with R

`github.com/endri81/ISLRv2-solutions`